

VANI M.
AI Engineer

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Professional Summary

AI Engineer with 4 years of hands-on experience in developing end-to-end AI pipelines, fine-tuning large language models (LLMs), and building intelligent data extraction and retrieval systems. Proficient in Python, SQL, Cypher and expertise in biomedical text processing, graph databases and knowledge graph construction.

Work Experience

Artificial Intelligence Engineer

January 2024 - Present

Nuve Software Labs Private Limited

- Designed and deployed AI pipelines using **Apache Airflow** for workflow automation.
- Implemented **Graph RAG** systems integrated with **Neo4j** for contextual knowledge retrieval.
- Developed **Named Entity Recognition (NER)** and **Relation Extraction (RE)** modules using **Large Language Models (LLMs)**.
- Built **Retrieval-Augmented Generation (RAG)** systems powered by **Elasticsearch** for semantic search.
- Fine-tuned **Phi-3** and **LLaMA-3** models for NER tasks using **PEFT** and **LoRA** optimization.
- Engineered solutions for **efficient unstructured data extraction** and representation.
- Developed an **agentic framework** using **CrewAI** for a **microlearning application**.
- Applied **PICO value extraction** using LLMs for the **biomedical domain**.

AI Developer

2023 May - 2023 December

MCMILLAN TECHNOLOGIES AND CONSULTANCY SERVICES Pvt. Ltd

- Designed a pipeline for data extraction from documents like invoices
- Designed a system for object detection
- Studied various possibilities for data extraction from documents

AI Trainee/ Developer

2021 November - 2023 April

ELEMENTZ ENGINEERS Pvt GUILD Ltd

- Assisted in the design and development of a chatbot solution using GPT-3 models, incorporating natural language processing and dialog management techniques.
- Assisted in the development and implementation of object detection algorithms for computer vision tasks.

Assistant Professor

2015 July - 2020 December

- Worked as Assistant Professor

Skills

Generative AI Frameworks & Tools: Hugging Face, LangChain, LlamaIndex, RAG, Fine-tuning, Ollama

LLM Models & Architectures: LLaMA, LLaVA, Crew AI, BERT, OpenAI

Databases & Search Systems: ElasticSearch, Neo4j, PostgreSQL, Chroma DB, FAISS

Workflow & Orchestration: Apache Airflow

Programming & Frameworks: Python, SQL, Cypher, FastAPI, Django (Basics)

Cloud Platforms: Paperspace, Azure, AWS, EC2

Education

MTech in Applied Electronics and Instrumentation

College of Engineering, Trivandrum

Post Graduated: 2015

CGPA: 9.08 / 10

BTech in Electronics & Communication

Govt. Rajiv Gandhi Institute Kottayam

Graduated: 2012

Percentage: 72.56%